

Dropshipping Inventory & Order Orchestrator

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: A valid order webhook results in supplier inventory verification and a supplier purchase order. Fail: A purchase order is created for an unavailable SKU.	Core functionality for automating dropshipping orders.	
FR-002	Functional	The system shall identify the supplier associated with each product in a customer order.	High	Pass: For every ordered SKU, the system displays or records the corresponding supplier. Fail: Any ordered SKU has no identified supplier when supplier information exists.	Required to route products to the correct supplier.	
FR-003	Functional	The system shall split a customer order into separate supplier orders when the order contains products from multiple suppliers.	High	Pass: An order containing products from two suppliers produces two separate supplier orders. Fail: Products belonging to different suppliers are sent in the same supplier purchase order.	Enables correct processing of multi-vendor orders.	
FR-004	Functional	The system shall update and store shipment tracking numbers received from supplier partners.	Medium	Pass: A valid tracking number received from a supplier is stored against the corresponding order. Fail: The tracking number is not associated with the correct order.	Allows the system and E-Store Owner to track shipments.	
FR-005	Functional	The system shall provide the E-Store Owner with the current status of customer orders and supplier fulfillment.	Medium	Pass: The E-Store Owner can view the current status of an order after supplier processing. Fail: The displayed status does not match the stored order status.	Helps the store owner monitor order fulfillment.	

NFR001	Non-Functional	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storm.	High	Pass: Benchmarking tests confirm that valid inbound webhooks receive HTTP 200 within 200 ms under simulated peak load. Fail: Response time exceeds 200 ms.	Prevents unnecessary webhook retries and supports reliable order processing.	
NFR002	Non-Functional	The system shall maintain order and tracking information without data loss during normal operation.	High	Pass: After a successful order or tracking update, the information remains available after system restart. Fail: Any successfully stored order or tracking information is lost.	Ensures reliable order and shipment tracking	